

spec

Feature Specification: A Repository That Clones, and a Fleet With No Unexplained Red

Feature Branch: main (this repository works on main only) **Created:** 2026-09-08 **Status:** Draft

Input: “Every red gate across the fleet must be resolved or converted into a stated, justified state — and the workshop repository must be cloneable on a second host. [...] Every fix must carry machine-produced deterministic evidence with three-valued gates, and no red may be closed by lowering a threshold, allow-listing a row, or re-baselining a count.”

Why this specification exists

A second engineer tried to clone the private curriculum repository and could not. That is a hard stop: work that cannot be obtained cannot be reviewed, run or contributed to, and every other quality property of that repository is unreachable behind it.

Around that sits a second problem of a different kind. **Eleven instruments across the fleet report red, and no reader can currently tell which reds are findings and which are the instruments working as designed.** A red that is correct and a red that is a defect look identical from the outside, and a list where most entries are permanent is a list nobody reads.

This specification treats those as one problem, because they have one remedy: every red must end in a state that a reader can act on — repaired, or declared with its evidence and its reason.

The measured starting state

#	Red	Measurement
R1	The repository cannot be reliably cloned	54 tracked archive part files totalling 2,624.8 MB ; pack 1.76 GiB . A second engineer's clone failed with curl 92 ... CANCEL (err 8), 9055 bytes of body are still expected, early EOF, invalid index-pack output. Root cause established by test, not inference — see <i>Root cause of R1</i> below. Only 68.1 MB across 2 blobs exists in history but not in the tree, so this is not history bloat.
R2	Continuation record stale	Commits changed watched governance files without updating the resumption record in the same commit. 7 pass · 1 drift · 15 notes.
R3	Hardcoded developer paths	3 files, 17 occurrences , plus 387 separately baselined. Two of the three are generated exports of one source document.
R4	Governance-source pin behind its remote	1 of 13 gitlinks differs from its remote. Difference is determined; direction is not ,

#	Red	Measurement
		because the remote commit is absent from the local object store.
R5	Provider-side triggering	1 confirmed standing trigger with no file-level remedy; 6 unverified rows on hosts with no read-only adapter.
R6	Module gate suite red	1 gate reports a violation and 4 could not run at all.
R7	25 areas carry no assessment	Not an evidence gap: 25 of 37 area documents were never sectioned into the passage registry, so no citation can resolve at their own source and no question can be authored.
R8	5 questions silently withheld	They cite redacted passages; the server withholds each one. They are dead in production and nothing says so to a learner.
R9	Interface paints 2 hue families	Against a floor of 6, measured on the served bundle by a corrected instrument whose three defects were fixed first — the count went down from 4 when the instrument became honest.
R10		

#	Red	Measurement
R11	3 documents fail publication review	62, 54 and 58 uncited claim blocks. They remain unpublished.
	Content-boundary gate red by design	~15,300 matches; a decision packet exists and ~1,540 rows are direction-measured as private-first and unread.

Root cause of R1 — established by controlled test, and a first attempt that was wrong

A refuted hypothesis is recorded here rather than deleted, because the wrong mechanism pointed the remedy in a different direction and the next reader should see why it was abandoned.

REFUTED: “GitHub’s HTTP/2 transport cancels on packs this large.”

This was inferred from the `curl 92` error text and from a blob-filtered clone completing in 2.01 s. It was reported as a finding without being tested. Two arms were then run from a third host against the same repository, authenticated over HTTPS, identical except for the negotiated protocol version:

arm	http.version	result
A	default (HTTP/2)	rc 0, 80 s, 2.6 GB
B	HTTP/1.1	rc 0, 82 s, 2.6 GB

A full HTTPS clone succeeds, over both protocol versions. HTTPS is not broken, HTTP/2 is not the cause, and “use SSH” is not a fix — it is a coincidence of which link happened to hold.

ESTABLISHED: git’s smart-HTTP transport cannot resume a failed fetch. A clone that dies at 90% restarts from zero. This repository therefore requires an **uninterrupted** 2.6 GB transfer to clone at all, and at that size the cumulative probability of at least one interruption somewhere in the network path is substantial on any marginal link. 9055 bytes of body are still expected is that interruption caught mid-body — a transfer too long to survive, not a protocol refusing a large pack.

This mechanism explains every observation; the refuted one explained two. It accounts for success over HTTPS here, success over SSH here, instant success blob-filtered, and failure there — with one cause.

What it changes. The remedy is unchanged: the target is the 2,624.8 MB, because size is what converts a transient blip into a total failure. What changes is the reasoning recorded in every artefact, and one design consequence: **a blob-filtered clone is worth documenting as ROBUST, not merely fast.** A 540 KB transfer that fails costs nothing to retry; a 2.6 GB one costs everything. FR-004's guard must therefore threshold on transfer size and state this mechanism in its own output, so the next reader does not re-derive the refuted one.

Clarifications

Session 2026-09-08

- Q: Where should recorded material live, given a 2,624.8 MB repository that HTTPS cannot clone? → A: **Git LFS for the archive parts, going forward.**
- Q: Are the 387 baselined hardcoded-path occurrences in scope, or do they stay declared? → A: **In scope — fix the 17 live ones AND work through the 387.**
- Q: Should the 6 provider rows on hosts with no read-only adapter get one? → A: **Write read-only adapters for both hosts.**
- Q: May the governance-source pin's direction be resolved, and the pin moved? → A: **Authorise the fetch, then fast-forward only if it is genuinely behind.**

What each decision obliges

LFS, and its honest limit. History rewriting is forbidden, so the existing blobs stay where they are: **a plain full clone remains large no matter what we do here.** What LFS changes is everything from now on, and it makes the documented default clone instant. SC-002's "under 250 MB to obtain" is therefore a claim about **the documented obtaining path**, not about the size of a naive `git clone` of full history — and the spec must say so rather than let the number imply something it cannot deliver. The existing archive tool already produces per-part and whole-file hashes, so integrity survives the move unchanged; that is what makes LFS viable here rather than a leap.

The 387 is a campaign, not a fix. The operator chose the larger option over the recommendation. Most of those occurrences sit **inside consumed submodules** and can only be repaired by upstream commits that return as gitlink bumps — so this spans repositories and cannot land in one change. Two obligations follow: the work must be **partitioned by owning repository** and reported per repository rather than as one number; and each row must first be **re-derived**, because a baseline that was never pruned overstates debt — a row fixed upstream and left in the baseline is counted as debt that no longer exists.

Adapters turn permanent unknowns into verdicts. Six rows have been rc 2 not because a check failed but because no instrument was ever registered for those hosts. Both expose public read-only APIs. The obligation: the adapters must report **three-valued** like every other probe, and an adapter that cannot reach its host must return 2 — replacing “never asked” with “asked and could not determine” is progress; replacing it with a green is not.

The pin fetch is authorised, and its sequence is fixed. Fetch to resolve direction, classify with `merge-base --is-ancestor` and a left/right count, then `merge --ff-only` — which refuses anything that is not a true fast-forward. The diff-stat must be filtered **by name** against every governance path before and after, because “no governance document changed” is a claim that must be measured rather than eyeballed. This pin has gone stale five times; the move closes an instance, not the class.

User Scenarios & Testing (*mandatory*)

User Story 1 - A second engineer obtains the repository (Priority: P1)

An engineer on a different machine clones the curriculum repository, follows its own written setup, and reaches a state its status command calls ready — without special flags, private knowledge, or a message to the author.

Why this priority: Nothing else in this specification is deliverable to anyone who cannot obtain the repository. It is the only red that blocks other people rather than degrading a signal.

Independent Test: Clone into an empty directory on a machine that has never held this work, run the documented bootstrap, and read the status output.

Acceptance Scenarios:

1. **Given** a machine with no prior copy, **When** the engineer clones by the documented command, **Then** the clone completes without a transport error.
 2. **Given** the clone, **When** they run the documented bootstrap, **Then** it obtains every dependency it needs, or names precisely what is missing and the exact command to obtain it.
 3. **Given** a chapter's recorded material, **When** they follow the documented step to obtain it, **Then** they get the recording, or a stated reason it is unavailable to them.
 4. **Given** the documented clone command, **When** a reader looks for it, **Then** it is in the repository's own front-door documentation, not in a chat message.
 5. **Given** the repository over time, **When** its size approaches the point where clones fail, **Then** something says so **before** a person discovers it by failing to clone.
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User Story 2 - A reader can tell a finding from a designed state (Priority: P1)

Someone runs the fleet's instruments and, for every red, can immediately tell whether it is a defect to fix, a known condition with a recorded reason, or a check that could not run — and what to do next in each case.

Why this priority: A list of eleven reds where most are permanent is a list nobody reads, and the reds that *are* defects then hide inside it. This is the difference between a gate suite that protects the work and one that decorates it.

Independent Test: Run every instrument, and for each red answer three questions from its output alone: is this a defect, what is the evidence, what happens next.

Acceptance Scenarios:

1. **Given** any instrument reporting red, **When** a reader reads its output, **Then** it states whether the red is a defect, a declared condition, or an unrunnable check.

2. **Given** a declared condition, **When** a reader reads it, **Then** it carries the evidence and the reason it is declared, and names who may lift it.
 3. **Given** a check that could not run, **When** it reports, **Then** it is distinguishable from both a pass and a failure, and it is **never** recorded as a pass.
 4. **Given** the full suite, **When** it is run, **Then** a single summary states how many reds are defects and how many are declared, and those two counts are never merged.
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User Story 3 - Every learner-facing defect is repaired or disclosed (Priority: P2)

A learner never encounters content that is silently broken. Where something is missing or withheld, they are told, rather than shown a shorter test or an absent section with no explanation.

Why this priority: These are user-visible defects, but they degrade rather than block, and they sit behind Story 1 for delivery.

Independent Test: Walk one area end to end and confirm nothing is silently absent — every omission is either repaired or stated.

Acceptance Scenarios:

1. **Given** a question whose supporting material is withheld, **When** the test is served, **Then** the question is repaired to cite available material, or removed with the removal recorded — never silently dropped from a learner's test.
 2. **Given** an area with no assessment, **When** a learner opens it, **Then** they are told it has none and why, rather than finding an absence.
 3. **Given** an area document citing withheld material as its primary reference, **When** it is published, **Then** its references resolve to material a reader can actually reach.
 4. **Given** the areas that could not carry a test for a structural reason, **When** that reason is removed, **Then** those areas become authorable and the coverage figure moves for a real cause.
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User Story 4 - The interface looks designed on the surfaces people use (Priority: P2)

A visitor perceives a varied, deliberate palette on the surfaces they spend their time looking at — not only on small indicators — while every pairing stays legible in both light and dark presentation.

Why this priority: It is the operator's twice-raised complaint and it is now measured. It changes how existing capability is perceived rather than what exists.

Independent Test: Sample what is painted on principal surfaces in both presentations; count hue variety and measure contrast within each pairing.

Acceptance Scenarios:

1. **Given** the served interface, **When** its principal surfaces are sampled, **Then** they carry materially more hue variety than a two-family scheme.
 2. **Given** any hue that is added, **When** a reader asks what it means, **Then** it carries a stated classification job — no hue is added only to raise a count.
 3. **Given** the measurement, **When** its population is defined, **Then** the population reflects the surfaces a visitor actually looks at, and any change to that population is justified independently of its effect on the count.
 4. **Given** both presentations, **When** contrast is measured, **Then** every text and non-text pairing meets its floor in each, measured separately.
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Edge Cases

- **A red is fixed by making the instrument weaker.** Forbidden and detectable: any change to a threshold, bucket, allow-list or baseline must be justified by a principle stated *before* the resulting count is known.
- **A repository shrinks below the failure threshold but keeps growing.** The size guard must warn on the trend, not only on the breach.
- **A recording is unavailable to a particular engineer.** Absence of permission is a stated reason, never a broken clone.

- **A check cannot run because a toolchain is missing.** That is could-not-determine, never a failure of the thing being checked, and never a pass.
- **The direction of a difference cannot be established read-only.** It is reported as determined-difference / undetermined-direction, and resolving it is an explicit decision, never an assumption.
- **A fix repairs the count but not the cause.** Each repaired red must name the mechanism that produced it, not only the symptom.
- **Two instruments disagree.** Both readings stand, with their populations stated; the disagreement is reported rather than resolved by preferring the convenient one.
- **A red is declared rather than fixed.** It must name who may lift the declaration and what evidence would.

Requirements (*mandatory*)

Obtainability

- **FR-001:** The repository **MUST** be obtainable by a documented command that completes without a transport error on a machine that has never held it.
- **FR-002:** Large recorded material **MUST NOT** be carried in the repository's own object store in a way that makes obtaining the repository fail.
- **FR-003:** The documented way to obtain recorded material **MUST** be in the repository's front-door documentation, and **MUST** state what a reader gets and what they may not.
- **FR-004:** An automated check **MUST** report the repository's obtainability cost and **MUST** warn **before** the point at which obtaining it fails, not after.
- **FR-005:** Existing published history **MUST NOT** be rewritten to achieve any of the above.

Red-state discipline

- **FR-006:** Every instrument reporting a non-clean result **MUST** state which of three states it is in: a defect, a declared condition, or a check that could not run.
- **FR-007:** A declared condition **MUST** carry its evidence, its reason, and who may lift it.
- **FR-008:** A check that could not run **MUST** be distinguishable from both a pass and a failure, and **MUST NEVER** be recorded as a pass.

- **FR-009:** A summary of the fleet MUST report defect count and declared-condition count separately and MUST NOT merge them.
- **FR-010:** No red may be closed by weakening the instrument. Any threshold, bucket, allow-list or baseline change MUST be justified by a principle stated independently of its effect on the resulting count.
- **FR-011:** Every fix MUST name the mechanism that produced the red, not only the symptom it removed.
- **FR-011a:** Debt spanning more than one repository MUST be partitioned by owning repository and reported per repository, never as a single number that hides where the work belongs.
- **FR-011b:** A baselined row MUST be re-derived before it is counted as debt. A row fixed upstream and left in a baseline overstates debt and must be pruned, not carried.
- **FR-011c:** An instrument added for a previously unprobed host MUST be three-valued, and MUST return could-not-determine when it cannot reach that host. Replacing “never asked” with “asked and could not determine” is progress; replacing it with a pass is not.

Learner-facing correctness

- **FR-012:** A question citing withheld material MUST be repaired to cite reachable material, or removed with the removal recorded. It MUST NOT remain silently withheld.
- **FR-013:** An area with no assessment MUST tell a learner so, with the reason.
- **FR-014:** A published document’s primary references MUST resolve to material a reader can reach.
- **FR-015:** The structural obstacle preventing areas from carrying assessments MUST be removed, and the resulting coverage figure MUST move for that cause and be reported.

Presentation

- **FR-016:** The served interface MUST carry materially more hue variety on its principal surfaces than a two-family scheme.
- **FR-017:** Every hue present MUST carry a stated classification job.
- **FR-018:** The measured population MUST reflect the surfaces a visitor actually uses, and any change to it MUST be justified independently of its effect on the count.
- **FR-019:** Contrast floors MUST hold for text and non-text pairings in both presentations, measured separately.

Evidence

- **FR-020:** Every requirement above **MUST** be covered by an automated check producing machine-readable evidence.
- **FR-021:** Every such check **MUST** be three-valued and **MUST** ship a paired demonstration that it detects the defect it exists to catch, driven by **data** rather than by altering the check.
- **FR-022:** Every check whose subject is a set **MUST** establish the set is non-empty before reporting the set contains no defects.
- **FR-023:** Any count reported about a corpus **MUST** carry evidence the corpus did not change during the measurement.

Key Entities

- **Red state** — one instrument's non-clean result. Has a kind (defect / declared / unrunnable), evidence, a reason, and for a declared state, who may lift it.
- **Obtainability budget** — the cost of obtaining the repository, with a warning threshold below the failure threshold.
- **Recorded material reference** — a pointer from the repository to material held outside it, with instructions and a stated availability condition.
- **Declared condition** — a red deliberately left standing, with its justification and lifting authority.

Success Criteria (*mandatory*)

Measurable Outcomes

- **SC-001:** An engineer on a machine that has never held the repository obtains it by the documented command, with **0** transport failures, in **3 of 3** attempts.
- **SC-002:** Obtaining the repository **by the documented command** costs **under 250 MB**, down from 2,624.8 MB. This is a claim about the documented path, not about a naive full-history clone: history rewriting is forbidden, so existing blobs remain and a full clone stays large. The spec states this rather than letting the number imply a reduction it cannot deliver.
- **SC-003:** **100%** of recorded material remains obtainable by a documented step, with **0** items lost.
- **SC-004:** **0** commits rewrite existing published history.

- **SC-005:** **100%** of non-clean instrument results state their kind; results whose kind cannot be determined from their output number **0**.
- **SC-006:** **0** instruments record a could-not-run as a pass.
- **SC-007:** **0** reds are closed by weakening an instrument; every threshold or baseline change carries a principle stated independently of the count it produces.
- **SC-008:** The count of unexplained reds — red, with no stated kind and no reason — reaches **0**, from a baseline of **11**.
- **SC-009:** **0** questions served to a learner cite material the server withholds, down from 5.
- **SC-010:** **100%** of areas without an assessment say so to a learner, with a reason.
- **SC-011:** The number of areas that cannot carry an assessment **for a structural reason** reaches **0**, from **25**; the number that carry none because the corpus does not support one is reported separately and honestly.
- **SC-012:** Principal surfaces carry at least **6** distinct hue families, up from **2**, and **100%** of hues present carry a stated job.
- **SC-013:** **100%** of text and non-text pairings meet their floor in both presentations; failures number **0**.
- **SC-014:** **100%** of requirements map to at least one automated check, and every such check ships a data-driven paired demonstration.
- **SC-015:** A reader answers “is this a defect, what is the evidence, what happens next” for every red from instrument output alone, without consulting a person.

Assumptions

- **Recorded material may live outside the repository.** No requirement here says the recordings must be inside it; only that they remain obtainable by a documented step.
- **Some reds are correct and will remain red.** Content-boundary detection, unreviewed documents and areas the corpus cannot support are expected to stay red; the requirement is that they are *declared*, not that they go green.
- **History rewriting is unavailable.** It is forbidden fleet-wide, so every obtainability fix must work forward from the current state.
- **Read-only probing is preferred.** Where a direction cannot be established without a mutating operation, the undetermined state is reported and the operation is a separate decision.
- **The audience is engineers and the operator**, not anonymous public users.

- **Two instruments measuring different populations may legitimately disagree**; the requirement is that populations are stated, not that they agree.

Dependencies

- A durable place to hold recorded material outside the repository's object store, reachable by the documented step.
- The existing archive-and-verify mechanism, which already produces per-part and whole-file hashes and refuses a corrupted part.
- The passage registry and its sectioning tool, for the structural obstacle in FR-015.
- The corrected served-surface measurement, for FR-016 to FR-019.

Out of Scope

- Rewriting published history, in any repository.
- Making the content-boundary detector green; only its rows' disposition is in scope.
- Authoring assessments for areas whose evidence does not support them.
- Changing provider-side settings that only an operator can change in a provider interface — measuring and reporting them is in scope; changing them is not.